

1. Morphological richness

ك ت ب (root)
كِتَاب *book* مَكْتَبَة *library*
كَاتِب *writer* مَكْتُوب *written*
one root, dozens of surface forms

2. Orthographic variation

إبن الهيثم
vs
ابن الهيثم
same person, two spellings of Alif

3. Diacritic sensitivity

علم *as written in the corpus*
عِلْم *knowledge*
عَلَم *flag*
عَلِم *he knew*
one spelling, three different words;
the diacritics that separate them are
omitted in almost all modern text

4. Diglossia

one question, six ways to ask it
MSA (the corpus) ما هذا؟
Sudanese شنو ده؟
Egyptian إيه ده؟
Levantine شو هاد؟
Gulf شنو هذا؟
Maghrebi أش هادا؟
the interrogative differs in every variety;
none of them matches the MSA ما

surface form and meaning do not correspond one to one

many forms for one concept (1, 2, 4) → the relevant passage is **missed**
one form for many concepts (3) → irrelevant passages are **matched**

★ remedy applied in this thesis: enrich the query *before* retrieval (Section 2.1.4)